

Cover letter

Dear Editors,

We submit "EcoNiche-Opt: a locked immune-ecology transcriptomic score for multicohort prediction of immune checkpoint blockade response" for consideration as a Research Article in Journal of Translational Medicine, with strongest relevance to the Disease Biomarkers, Cancer Microenvironment, and Data-driven Clinical Decision Processes sections.

The study addresses a translational barrier in immune-checkpoint blockade biomarker development: public transcriptomic cohorts contain useful immune signals, but response labels, sampling status, assay platforms, endpoint definitions, and tumour-immune ecological states are often heterogeneous. EcoNiche-Opt addresses this gap by combining auditable cohort and response-label curation, endpoint-stratified evaluation, signed-rank ecological module scoring, module-interaction edges, biologically constrained optimization, leave-one-dataset-out testing, locked external validation, and a reproducible 62-gene panel-compatible scoring rule.

In primary melanoma benchmarks, EcoNiche-Opt showed FDR-supported improvement over a predeclared eight-signature family. Locked external and panel-transfer analyses showed reproducible portability without model refitting, and single-cell localization plus ecological-edge analyses linked the score to antigen presentation, T/NK effector activity, T-cell dysfunction, myeloid suppression, and stromal exclusion. The work therefore provides not only a predictor, but a reusable translational framework for developing and validating immune-ecology biomarkers across heterogeneous ICB cohorts.

All authors have approved the manuscript for submission. The manuscript has not been published and is not under consideration elsewhere. The authors declare no competing interests. The study used previously published de-identified public or controlled-access data and did not recruit new human participants or collect new biospecimens.

Sincerely,

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